Model glider wing

Request: a wing for a 1.2 kg model glider, at most 2 m span

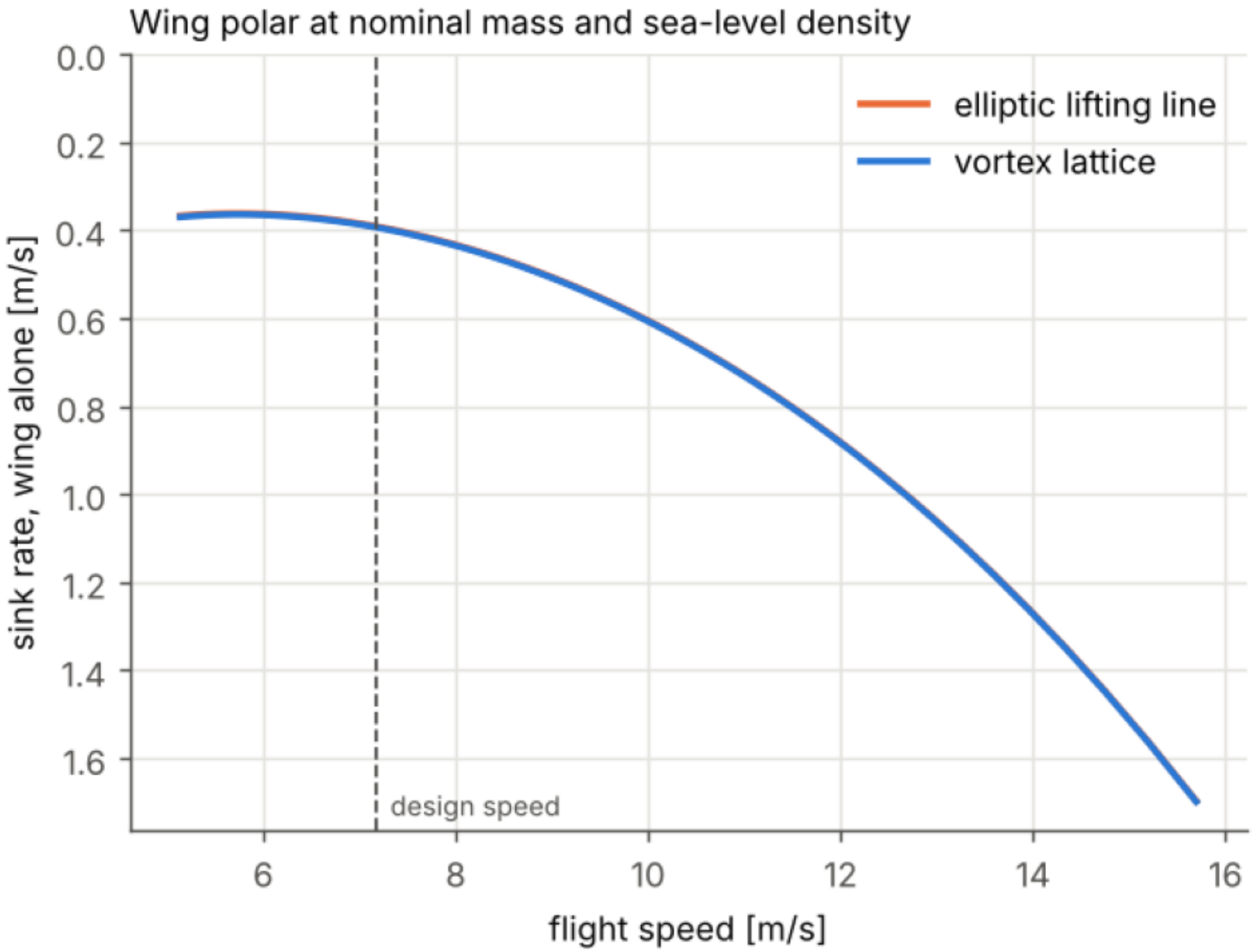
Verdict: will work   P = 1.000 (95 % CI 0.998-1.000)   trust: APPROVED

Design:
   span: 2 m
   aspect ratio: 7.693
   taper ratio (tip / root chord): 0.3254
   section: NACA 2412

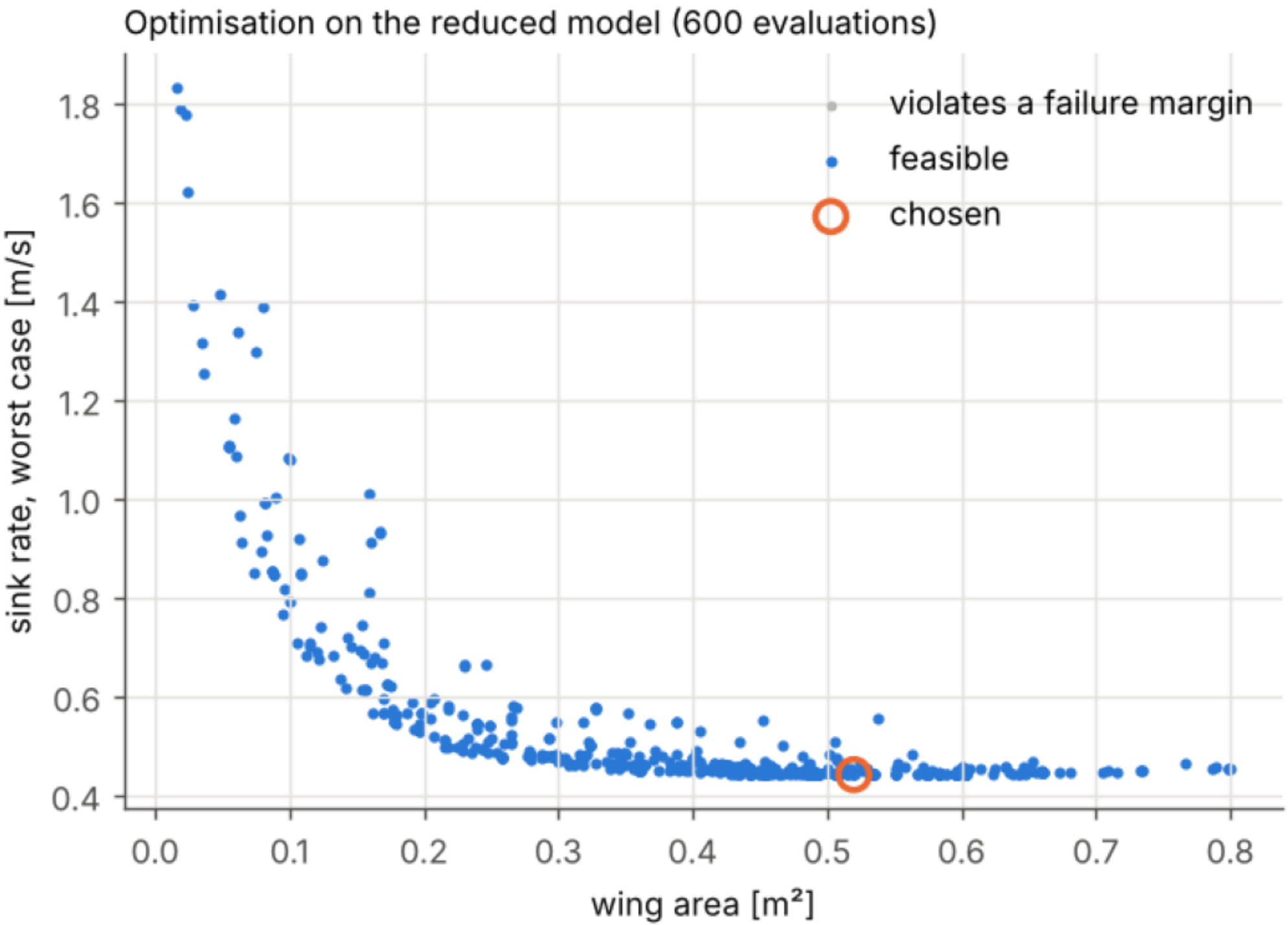
Numbers (fidelity):
   ld: 16.3 - [vortex lattice, coarse]
   lift coefficient below the design limit: 0   [vortex lattice, coarse]
   cl: 0.8 - [vortex lattice, coarse]
   Reynolds number above the critical value: 0.9321   [vortex lattice, coarse]
   span_eff: 0.9903 - [vortex lattice, coarse]
   sink_m_s: 0.4406 m/s [vortex lattice, coarse]
   area_m2: 0.52 m² [vortex lattice, coarse]
   re_mac: 1.352e+05 - [vortex lattice, coarse]
   speed_m_s: 7.183 m/s [vortex lattice, coarse]
   lift coefficient below the design limit: 0   [elliptic lifting line]
   Reynolds number above the critical value: 0.9321   [elliptic lifting line]
   sink_m_s: 0.4409 m/s [vortex lattice, fine]
   span_eff: 0.9894 - [vortex lattice, fine]
   speed_m_s: 7.183 m/s [vortex lattice, fine]
   ld: 16.29 - [vortex lattice, fine]
   margin_cl: 0   [vortex lattice, fine]
   margin_re: 0.9321   [vortex lattice, fine]
   cl_alpha: 4.681   [vortex lattice, fine]

Verification:
   sink_m_s: full 0.4409, reduced 0.4406, rel. err 0.000492
   span_eff: full 0.9894, reduced 0.9903, rel. err 0.0009035

Limits:
   - Wing only: fuselage, tail and interference drag are not included.
   - Inviscid lift (vortex lattice): no stall prediction; the design lift coefficient is capped instead.
   - Profile drag from a flat-plate correlation, not from a viscous solver.
   - No structural sizing (spar, launch loads).



Design speed chosen at the worst corner: mass x1.05, density x0.94, profile drag x1.25


Optimisation on the reduced model (600 evaluations)
violates a failure margin
feasible
chosen
sink rate, worst case [m/s]
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
wing area [m²]

